

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer..

Problem 1. (5 points) Find a general solution using the substitution method for the differential equation: $y' = (x+y)^2$.

$$\text{let } \underline{V = x+y}, \text{ then } y = V - x \Rightarrow \frac{dy}{dx} = \frac{dV}{dx} - 1$$

thus we get

$$\frac{dV}{dx} - 1 = V^2$$

$$\frac{dV}{dx} = V^2 + 1$$

$$\frac{1}{V^2+1} dV = dx$$

$$\int \frac{1}{V^2+1} dV = \int dx$$

$$\arctan(V) = x + C$$

$$V = \tan(x+C)$$

$$x+y = \tan(x+C)$$

$$y(x) = \tan(x+C) - x$$

Problem 2. (5 points) Verify whether the given differential equation is exact and then solve it if yes: $(4x - y) dx + (6y - x) dy = 0$.

$$M(x, y) = 4x - y, \quad N(x, y) = 6y - x$$

$$\Rightarrow \frac{\partial M}{\partial y} = -1, \quad \frac{\partial N}{\partial x} = -1 \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \text{ i.e. exact DE!}$$

We know the solution has the form of $\underline{F(x, y) = C}$. ($\frac{\partial F}{\partial x} = M, \frac{\partial F}{\partial y} = N$)

$$\begin{aligned} F(x, y) &= \int M(x, y) dx + g(y) \\ &= \int (4x - y) dx + g(y) = 2x^2 - xy + g(y) \end{aligned}$$

$$\begin{aligned} \frac{\partial F}{\partial y} &= N \Rightarrow -x + g'(y) = 6y - x \Rightarrow g'(y) = 6y \\ &\Rightarrow g(y) = \int 6y dy = 3y^2 + C_1 \end{aligned}$$

$$\text{Thus } F(x, y) = 2x^2 - xy + 3y^2 + C_1$$

$$\Rightarrow \text{the solution is } 2x^2 - xy + 3y^2 = C$$